DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	:	S-ITCS123LA
COURSE TITLE	:	Intermediate Programming
COURSE TYPE	:	Laboratory
COURSE CREDIT	:	one unit
LEARNING MODALITY	:	Fully Onsite
PRE-REQUISITES	:	S-ITCS112/S-ITCS112LA
CO-REQUISITES	:	S-ITCS123
PROFESSOR	:	ROCHELLE D. SERMAÑA
CONSULTATION HOURS	:	MONDAY : 1300-1500 (ONLINE) TUESDAY: 0700-0900 (COS100-C) THURSDAY: 1300-1400(COS100-C) 1400-1600 (ONLINE)

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VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

Includes review of control structures and data types with emphasis on structured data types. Application of object-oriented programming paradigm, focusing on the definition and use of classes along with the fundamentals of object-oriented design is also part of the discussion as well as basic analysis of algorithms, searching and sorting techniques.

COURSE LEARNING OUTCOMES:

By the end of this course, students are expected to:

CLO1. understand programming concepts;
CLO2. implement file processing;
CLO3. examine program functionality;
CLO4. develop cleaner, more efficient and more manageable programs in solving problems;
CLO5. develop graphical user interfaces (GUI) for Java programs;
CLO6. evaluate searching and sorting algorithms;

CLO7. apply Java's object oriented programming concepts in the writing and editing of various levels of applications; and
CLO8. demonstrate 21st century skills in all learning activities.

LEARNING PLAN:
Learning Outcomes for Midterm Period

CLO1	Understand programming concepts	TLO1	Review basic control structures
		TLO2	Understand syntax and semantics of new programming constructs
		TLO3	Differentiate structured and unstructured data
		TLO4	Discuss how data is stored and its use
		TLO5	Explain operations that can be performed on a given structure
		TLO6	Observe how file processing works.
CLO2	Implement file processing	TLO7	Read data from a file
		TLO8	Write data to a file
		TLO9	Perform operations on a file
CLO3	Examine program functionality	TLO10	Evaluate program execution by tracing program output.
		TLO11	Test correctness of programs against several test data
		TLO12	Validate input of the program
CLO4	Develop cleaner, more efficient and more manageable programs in solving problems	TLO13	Choose an appropriate data and programming structure
		TLO14	Construct a program to implement specific searching and sorting algorithm
		TLO15	Construct user-defined methods
		TLO16	Construct recursive methods
CLO5	Develop graphical user interfaces (GUI) for Java programs.	TLO17	Design GUIs
		TLO18	Configure GUI properties
		TLO19	Perform event-driven programming
CLO8	Demonstrate 21st century skills in all learning activities.	TLO29	*Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real- World Problem- Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Allotted Hours
Module 0	Week 1 Jan 20-25	• Gospel, Reflection and Prayer • Presentation of the Course Syllabus • Module 0: Course Introduction	Schoolbook PowerPoint	3
Module 1 CLO1 CLO3 CLO4 CLO5 TLO1 TLO2 TLO13 TLO17 TLO19	Week 2 Jan 27-Feb 1	• Gospel, Reflection and Prayer • Program Demonstration of Module 1 : Review of Control Structures, Data Types and Structured Data Types • Enabling Assessment 1: Lab Exercise on Programming Problems from Module 1	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	3
Module 2 CLO1 CLO2 TLO2 TLO6 TLO10 TLO13	Week 3 Feb 3 - 8	• Gospel, Reflection and Prayer • Program Demonstration of Module 2: • Review of Control Structures, Data Types and Structured Data Types Part 2 • Enabling Assessment 2: Lab Exercise on Program Simulation of Module 2 Part 2	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	3

	Week 4 Feb 10-15	• Gospel, Reflection and Prayer • Program Demonstration of the Module 2: File Processing • Enabling assessment 3: Lab Exercise on Programming Problems of Module 2	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	3
Module 3 CLO1 CLO3 CLO5 TLO2 TLO13 TLO15 TLO17 TLO19	Week 5 Feb 17-22	• Gospel, Reflection and Prayer • Program Demonstration of Module 3: User-Defined Methods and Recursive Methods • Enabling assessment 4: Lab Exercise on Programming Concepts of Module 3	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	3
	Week 6 Feb 24-Mar 1	• Gospel, Reflection and Prayer • Enabling assessment 5: Lab Exercise On Program Analysis of Module 3	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	3
	Week 7 Mar 3-8	• Gospel, Reflection and Prayer • Enabling assessment 6: Lab Exercise On Programming Concepts of Module 3	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	3
Module 1 to 3	Week 8 Mar 10-15	***Summative Assessment:*** ***MIDTERM LAB EXAM*** Application Type covering Module 1, 2, 3		3

TOTAL	*24*

Learning Outcomes for Final Period
Learning Outcomes for Final Period

CLO1	Understand programming concepts	TLO1	Review basic control structures
		TLO2	Understand syntax and semantics of new programming constructs
		TLO3	Differentiate structured and unstructured data
		TLO4	Discuss how data is stored and its use
		TLO5	Explain operations that can be performed on a given structure
		TLO6	Observe how file processing works.
CLO3	Examine program functionality	TLO10	Evaluate program execution by tracing program output.
		TLO11	Test correctness of programs against several test data
		TLO12	Validate input of the program
CLO4	Develop cleaner, more efficient and more manageable programs in solving problems	TLO13	Choose an appropriate data and programming structure
		TLO14	Construct a program to implement specific searching and sorting algorithm
		TLO15	Construct user-defined methods
		TLO16	Construct recursive methods
CLO5	Develop graphical user interfaces (GUI) for Java programs.	TLO17	Design GUIs
		TLO18	Configure GUI properties
		TLO19	Perform event-driven programming
CLO6	Evaluate searching and sorting algorithms	TLO20	Explain how searching and sorting algorithm works
		TLO21	Compare and contrast the different searching and sorting algorithms
		TLO22	Decide which algorithm to use in a specific problem
CLO7	Apply Java's object oriented programming concepts in the writing and editing of various levels of applications.	TLO23	Create user-defined classes
		TLO24	Create static and non-static data and methods
		TLO25	Apply access modifiers
		TLO26	Implement UML Diagrams in designing user-defined classes
		TLO27	Create array of classes
CLO8	Demonstrate 21st century skills in all learning activities.	TLO29	*Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real- World Problem- Solving, Use of ICT*

			(not part of topic, but integrated in assessments)

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Allotted Hours
Module 4 CLO1 CLO3 CLO5 CLO7 TLO2 TLO10 TLO13 TLO15 TLO17 TLO19 TLO23 TLO27	Week 10 Mar 24-29	• Gospel, Reflection and Prayer • Program Demonstration of Module 4: 4: User Defined Classes Part 1 • Enabling Assessment 1: Laboratory Exercise on Programming Concepts of Module 4 **SUSTAINABILITY WEEK**	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	3
	Week 11 Mar 31- Apr 5	• Gospel, Reflection and Prayer • Program Demonstration of Module 4: User Defined Classes Part 2 • Enabling assessment 2: Laboratory Exercise on Program Simulation of Module 4	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	3

	Week 12 Apr 7-12	• Gospel, Reflection and Prayer • Enabling assessment 3: Laboratory Exercise on Programming Concepts of Module 4	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	3
	Week 13 Apr 14-19	***Independent Learning Days (April 14-15) Holy Week (April 16-19)*** • Program Demonstration on Module 4: User Defined Classes Part 2 (Continuation)		3
Module 5 CLO7 TLO7 CLO8 TLO8	Week 14 Apr 21-26	***Research Week*** • Program Demonstration on Module 5 : Searching Algorithms • Enabling assessment 4: Laboratory Exercise on Program Simulation of Module 5		3
Module 6 CLO1 CLO3 CLO6 TLO2 TLO10 TLO13 TLO22	Week 15 Apr 28- May 3	• Gospel, Reflection and Prayer • Program Demonstration of the Module 6 : Sorting Algorithms • Enabling assessment 5: Laboratory Exercise on Program Analysis of Module 6	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	3

	Week 16 May 5-10	• Gospel, Reflection and Prayer • Enabling assessment 6: Lab Exercise on Program Analysis of Module 6	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	3
Module 4 to 6	Week 17 May 12-17	***Summative Assessment:*** ***FINAL LAB EXAM*** Application Type (Output-Based) covering Module 4,5,6		3
TOTAL				***24***

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	40%	Enabling Assessments	40%
Summative Assessment (Midterm Examination)	60%	Summative Assessment	60%
	100%		100%

Overall Final Grade = $\frac{\text{Midterm + Final}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Onsite Learning Mode. All activities will be done in the laboratory classroom and field.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar.

3. Laboratory Policies
    a. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    b. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones also cannot be used as calculators during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance

of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

4. Communication and Feedback
    a. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    b. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

5. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

6. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

7. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.

REFERENCES

Onsite References

Provider	Reference Material
AEA Lib QA 76.73.J38 .Sch35 2022	Schildt, H .2022. Java : a beginner's guide.(New York): McGraw-Hill Education

Online References

Call number or e-provider	Reference Material
eBook	Prudhomme, G. (2020). *Java Programming Applications*. Arcler Press.
Web Resource	Java Classes and Objects. c1999-2024. Refsnes Data. [accessed 2025 Jan 29]. https://www.w3schools.com/java/java_classes.asp
Web Resource	Searching Algorithms.GeeksforGeeks.Sanchhaya Education Private Limited. [accessed: 2025 Jan 29].https://www.geeksforgeeks.org/searching-algorithms/
Web Resource	Sorting Algorithms.GeeksforGeeks.Sanchhaya Education Private Limited. [accessed: 2025 Jan 29].https://www.geeksforgeeks.org/sorting-algorithms/

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